Sergo Tovmasyan | ML Engineer/Researcher

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Profile

Highly motivated AI researcher and ML engineer with a strong foundation in physics and mathematics, preparing to enter the YSU Faculty of Physics. I am eager to apply my practical experience in machine learning, computer vision, and deep learning to national technological initiatives and public administration. Equipped with hands-on experience in building end-to-end AI pipelines, combined with active legislative and civic experience in the Armenian Youth Parliament, I am committed to advancing Armenia's AI ecosystem and integrating innovative solutions into the public sector. (Military deferment ensures uninterrupted focus for the next four years).

Core Skills

- AI & Technical: Python, Java, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Transformers, Diffusion models, Computer Vision.
- Public Policy & Civic: Legislative Drafting, Public Policy Analysis, Public Speaking, Project Management, Civic Leadership.
- Other: Mathematics, Physics, Algorithms in AI, Armenian, English, Russian.

Mentorship & Research

● Advanced Mathematics & ML Research: Actively studying higher mathematics and machine learning architectures under the mentorship of Dr. Tigran Ishkhanyan (Senior Research Scientist, Wolfram Research), Grick Tadevosyan (PhD Student, MBZUAI), and Smbat Gogyan.
● ANOAI & IOAI Competitions: Reached the 3rd national stage and placed 11th overall in the Armenian team selection. Actively preparing for IOAI 2026 as a selected pre-team member.

Public Sector & Civic Experience

- Member of the Youth Parliament 2026 | National Assembly of the Republic of Armenia (Aug 2026) - Served on the Standing Committee on Territorial Administration, Local Self-Government, Agriculture and Environment.
  Co-authored and drafted amendments to the RA Law "On Local Self-Government".
  Successfully presented and defended the legislative draft from the National Assembly podium, demonstrating strong public speaking and policy analysis skills.
- Project Coordinator | "Hayakert" Youth Initiative (2024) - Initiated and coordinated community youth projects in Vanadzor under the YIC "Youth in Action" program, showcasing strong organizational leadership.
- Co-leader | Eco-classrom project(2022) - Co-led a team project that achieved first place at the Armenian Tree Project (ATP) summit.

Career Summary

September 2024 - Present

1. Omnicode AI - AI/ML Engineer Intern: Contributed to machine learning and artificial intelligence projects, gaining practical industry experience in developing and optimizing AI solutions.
2. MesRop AI (Armenian Handwritten Text Recognition): Designed a specialized OCR system using CNNs and fine-tuned Vision Transformers. Built the dataset from scratch and achieved 98% accuracy on the test set.
3. SmartEYE (Assistive AI for the Visually Impaired): Contributed to an AI-powered assistive system designed to help visually impaired users understand their surroundings and navigate safely. Integrated YOLO and LLM tools to optimize spatial analysis.
4. Multi-Language OCR Application: Deployed a PyTorch-based model within a client-server pipeline using Flask as a final project for Samsung Innovation Campus. Integrated an Android (Java) interface and Firebase.
5. Armigo (AI Trip Planner): Contributed to an AI-driven mobile app for collaborative travel planning, featuring chat-based interaction and automated route generation.

Education & Qualifications

➢ Yerevan State University (YSU) - Faculty of Physics
➢ Vanadzor Evrika High School and Generation AI program - 2026
➢ National Assembly Summer School and Youth Parliament (2025/2026)
➢ Samsung Innovation Campus (2025) & 1991 Academy ML Course (2025)

Hobbies and Interests

➔ Kaggle & Hackathons: Active participant in hackathons and currently competing in a NEOAI Kaggle competition.
➔ Scientific Interests: Strong curiosity in interdisciplinary sciences, particularly biology. Expanded focus into quantum physics, alongside an active interest in historical and philosophical developments.
➔ Civic Leadership & Networking: Passionate about networking and actively engaged in civic initiatives, public debates, and youth forums;